



# Bilateral trade

**Model: Symmetric IPV** plus regularity (i.e, increasing  $v_i(\cdot)$ )

- Classic auctions with appropriate reserve price maximize seller's expected revenue.
- *When maximizing expected seller's revenue, it is fitting to consider the seller as the mechanism designer and to assume that the seller knows her valuation.*
- Classic auctions without reserve price maximize social surplus provided the seller's valuation is known.

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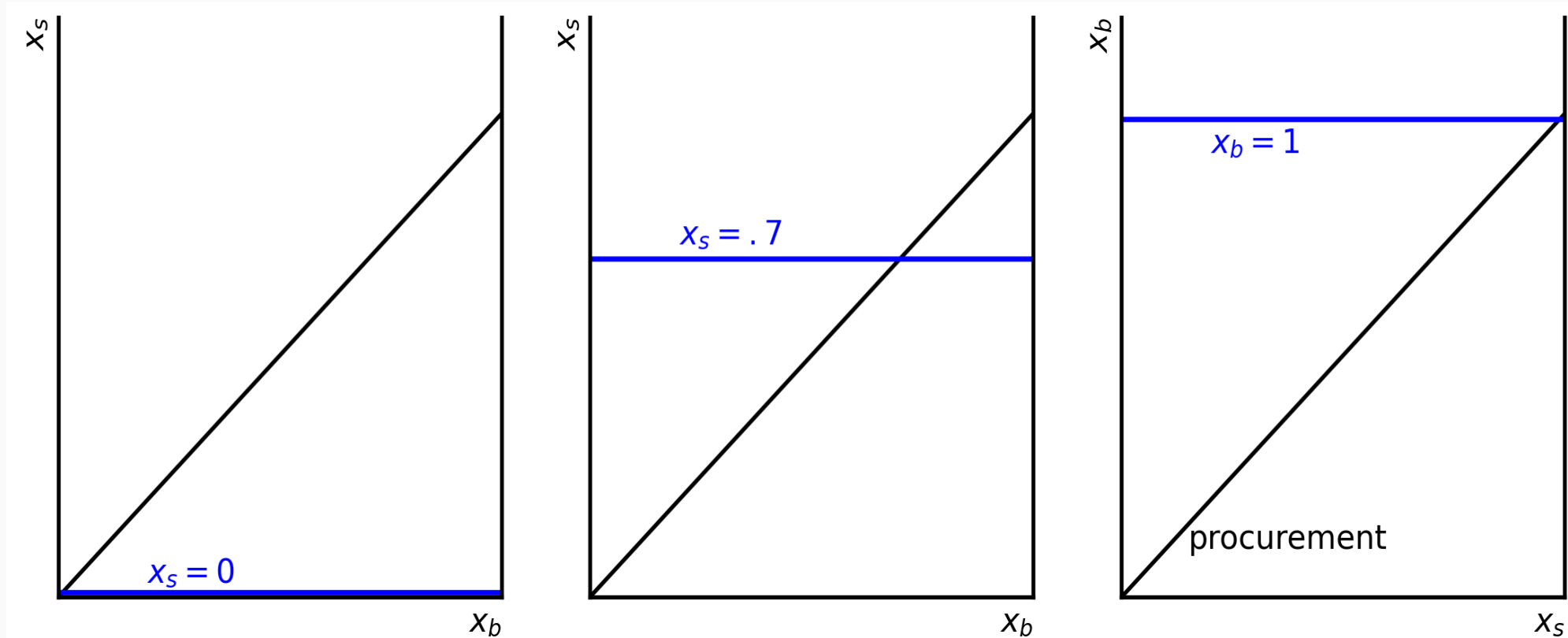
## Model: IPVVM without symmetry or regularity

- Without symmetry or regularity results are “fewer.”
- The revenue maximizing DRM is known (Myerson (1981)). It is not obvious how to use it. The economic implications are not as developed.

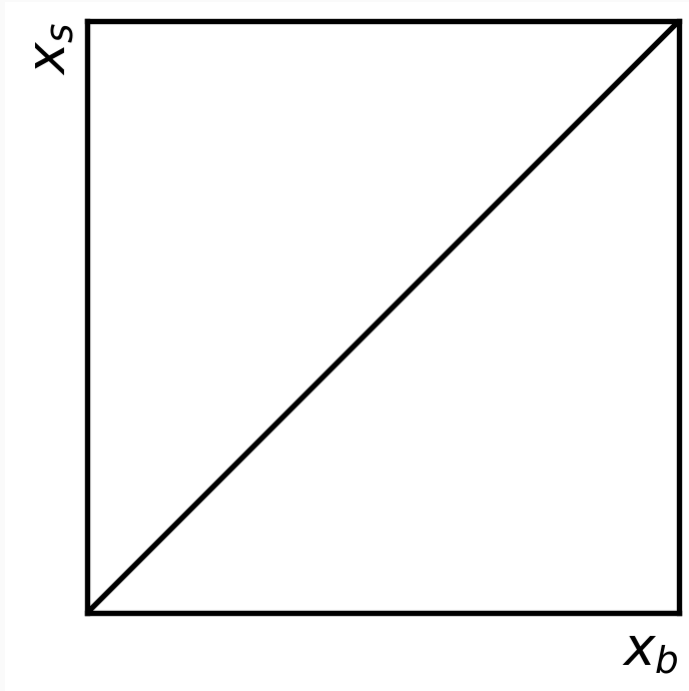
# Market design, efficiency of bilateral trade

- Are trading mechanisms “efficient” in the sense that trade takes place always if the buyer’s valuation exceeds the seller’s valuation and never otherwise?
- Appropriate model:
  - Seller and buyer have private information (types);
  - the mechanism designer does not observe agents’ types.
- IPVVM variation: a buyer and a seller with private information,  $x_b, x_s$  respectively, r.v. in  $[c, d]$ .
- Question. Is there an ex post efficient mechanism? That is, is there a mechanism that ensures that trade takes place if and only if  $x_b > x_s$ ?

# Fixed uninformed agent's valuation



# Bilateral private information



- Both buyer and seller observe their valuations  $x_b$  and  $x_s$  distributed according to pdfs following  $f_b(x_b)$  and  $f_s(x_s)$  respectively.

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- See Vickrey (1961). There is no mechanism with the following properties:
    - i. truthtelling is a DS equilibrium;
    - ii. no outside subsidies;
    - iii. ex post Pareto efficiency.
  - Both d'Aspremont and Gérard-Varet (1979) and Arrow (1979) find mechanism with
    - i. Bayesian IC;
    - ii. no outside subsidies but no IR constraints,
    - iii. ex post efficiency.
  - Myerson and Satterthwaite (1983) prove that no mechanism satisfies (i) Bayesian IC; (ii) IR and no outside subsidies; (iii) ex post efficiency.

# Bilateral trade–Theorem

**Theorem 1 (Myerson and Satterthwaite)** IPV with 2 agents, a buyer and a seller, with valuations  $x_b \in [c, d]$  and  $x_s \in [c, d]$  respectively.

(Valuations are independently distributed, not necessarily identically.)

There is no mechanism that is (i) Bayesian IC; (ii) satisfies IR (without outside resources); and (iii) is efficient.

- The supports need to overlap but do not need to coincide.

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*Proof.*

- Let  $(\tilde{p}, \tilde{t})$  be a BIC mechanism where

$$(\tilde{p}, \tilde{t}) = (p_b(x_b, x_s), t_b(x_b, x_s), p_s(x_b, x_s), t_s(x_b, x_s))$$

with  $p_b(x_b, x_s) + p_s(x_b, x_s) = 1$  and  $t_b(x_b, x_s) + t_s(x_b, x_s) = 0$ .

- $(\tilde{p}, \tilde{t})$  may be described by two functions instead of four,

- $p : [c, d]^2 \rightarrow [0, 1]$ , and  $t : [c, d]^2 \rightarrow \mathbb{R}$  where

- $p_b = p$  (probability that buyer gets the object),
- $p_s = 1 - p_b$  (probability seller keeps object),
- $t_b = t$  (payment from buyer to seller),  
 $t_s = -t_b$  (amount received by seller).

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Then,

$$u_b(x_b, x_s) = p(x_b, x_s)x_b - t(x_b, x_s)$$

$$u_s(x_b, x_s) = t(x_b, x_s) - p(x_b, x_s)x_s$$

$$U_b(x_b) = E_{x_s}[u_b(x_b, x_s)] = Q_b(x_b)x_b - T_b(x_b)$$

$$U_s(x_s) = E_{x_b}[u_s(x_b, x_s)] = T_s(x_s) - Q_s(x_s)x_s$$

where

$$T_s(x_s) = E_{x_b}t(x_b, x_s)$$

$$Q_s(x_s) = E_{x_b}p(x_b, x_s)$$

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The mechanism must satisfy two constraints

- Interim individual rationality

$$(\forall x_b) U_b(x_b) \geq 0 \text{ and } (\forall x_s) U_s(x_s) \geq 0.$$

- Bayesian incentive compatibility (BIC),

$$\begin{aligned} U_s(x_s) &= U_s(d) + \int_{x_s}^d Q_s(y) \, dy \\ U_b(x_b) &= U_b(c) + \int_c^{x_b} Q_b(y) \, dy \end{aligned} \tag{1}$$

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When players observed  $(x_b, x_s)$ , total interim surplus

$$U_s(x_s) + U_b(x_b) = Q_b(x_b)x_b - T_b(x_b) + T_s(x_s) - Q_s(x_s)x_s.$$

Since  $-E_{x_b}T_b(x_b) + E_{x_s}T_s(x_s) = 0$ , expected total surplus is

$$\begin{aligned} E[U_s(x_s) + U_b(x_b)] &= E[Q_b(x_b)x_b - Q_s(x_s)x_s] \\ &= \int_c^d \int_c^d [Q_b(x_b)x_b - Q_s(x_s)x_s] f_b(x_b)f_s(x_s) dx_b dx_s. \end{aligned}$$

$$\begin{aligned} E[U_s(x_s) + U_b(x_b)] \\ &= \int_c^d \int_c^d p(x_b, x_s)[x_b - x_s] f_b(x_b)f_s(x_s) dx_b dx_s \end{aligned} \tag{2}$$

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From Equation 1 (IC, integrability condition)

$$\begin{aligned} E[U_s(x_s) + U_b(x_b)] &= U_s(d) \\ &+ \int_c^d \left[ \int_{x_s}^d Q_s(y) \, dy \right] f_s(x_s) \, dx_s \\ &+ U_b(c) + \int_c^d \left[ \int_c^{x_b} Q_b(z) \, dz \right] f_b(x_b) \, dx_b \end{aligned}$$

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Integrating by parts both terms,

$$\begin{aligned} & \int_c^d \left[ \int_c^{x_b} Q_b(z) \, dz \right] f_b(x_b) \, dx_b \\ &= \int_c^d Q_b(x_b) \, dx_b - \int_c^d F_b(x_b) Q_b(x_b) \, dx_b \\ &= \int_c^d [1 - F_b(x_b)] Q_b(x_b) \, dx_b \end{aligned}$$

and

$$\int_c^d \left[ \int_{x_s}^d Q_s(y) \, dy \right] f_s(x_s) \, dx_s = \int_c^d F_s(x_s) Q_s(x_s) \, dx_s$$

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$$E[U_s(x_s) + U_b(x_b)] = U_s(d) + U_b(c) + \int_c^d \int_c^d \{F_s(x_s)f_b(x_b) + [1 - F_b(x_b)]f(x_s)\}p(x_b, x_s) dx_b dx_s$$

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Equating the two expressions of total surplus, Equation 3 and Equation 2, and solving, we obtain

$$\begin{aligned} U_s(d) + U_b(c) &= \\ &= \int_c^d \int_c^d \left\{ \left[ x_b - \frac{1 - F_b(x_b)}{f_b(x_b)} \right] - \left[ x_s + \frac{F_s(x_s)}{f_s(x_s)} \right] \right\} \\ &\quad p(x_b, x_s) f_s f_b \, dx_b \, dx_s \\ &\geq 0 \end{aligned}$$

Where the inequality follows by IR. Suppose

$$p(x_b, x_s) = \begin{cases} 1 & \text{if } x_b \geq x_s \\ 0 & \text{otherwise} \end{cases}$$

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$$\begin{aligned}
& \int_c^d \int_c^d \left\{ \left[ x_b - \frac{1 - F_b(x_b)}{f_b(x_b)} \right] - \left[ x_s + \frac{F_s(x_s)}{f_s(x_s)} \right] \right\} p(x_b, x_s) \\
& \qquad \qquad \qquad f_s \, dx_s f_b \, dx_b \\
& = \int_c^d \int_c^{x_b \wedge d} \left\{ \left[ x_b - \frac{1 - F_b(x_b)}{f_b(x_b)} \right] - x_s \right\} f_s - F_s(x_s) \Big\} f_b \, dx_s \, dx_b
\end{aligned}$$

where the RHS uses the definition of  $p(x_b, x_s)$ .

$$\begin{aligned}
&= \int_c^d \left[ \left( x_b - \frac{1 - F_b(x_b)}{f_b(x_b)} - x_s \right) F_s(x_s) \right]_c^{(x_b \wedge d)} f_b \, dx_b \\
&= \int_c^d \left[ \left( x_b - \frac{1 - F_b(x_b)}{f_b(x_b)} - (x_b \wedge d) \right) F_s(x_b \wedge d) \right] f_b \, dx_b \\
&= \int_c^d [-[1 - F_b(x_b)] F_s(x_b)] \, dx_b \\
&\quad + \int_c^d [(x_b - d) f_b(x_b) + (F_b(x_b) - 1)] \, dx_b \\
&\leq 0
\end{aligned}$$

QED

# Alternative proof

This proof is due to Williams (1999).

1. Consider a DRM that satisfies BIC and ex post efficiency.

- If  $x_s < x_b$  then
  - Buyer pays  $x_s$  to seller and gets object.
  - Seller gives up object and receives  $x_b$ .
- If  $x_s > x_b$  then
  - there is no transaction and no transfers.

Note it is a DS to report valuation truthfully.

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2. Corresponding expected transfers are

- Buyer pays  $T_b(x_b) = E_{x_s}[x_s \chi_{x_s < x_b}(x_s)]$ .
- Seller receives  $T_s(x_s) = E_{x_b}[x_b \chi_{x_s < x_b}(x_b)]$ .

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- Let  $(p', t')$ , be any mechanism that is ex post efficient. By the Expected Transfer Theorem, it must have the same expected transfers as the proposed mechanism up to a constant:
    - Buyer pays  $T'_b(x_b) = E_{x_s}[x_s \chi_{x_s < x_b}(x_s)] + b$ .
    - Seller receives  $T'_s(x_s) = E_{x_b}[x_b \chi_{x_s < x_b}(x_b)] + s$ .

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3. Players expected payoffs in mechanism  $(p', t')$ .

$$\begin{aligned} U'_b(x_b) &= Q'_b(x_b) \cdot x_b - T'_b(x_b) \\ &= \overbrace{E_{x_s}[x_b \chi_{x_s < x_b}]} - \overbrace{E_{x_s}[x_s \chi_{x_s < x_b}(x_s)]} + b \\ &= E_{x_s}[(x_b - x_s) \chi_{x_s < x_b}] + b \end{aligned}$$

$$\begin{aligned} U'_s(x_s) &= T'_s(x_s) - Q'_s(x_s) \cdot x_s \\ &= \overbrace{E_{x_b}[x_b \chi_{x_s < x_b}]} + s - \overbrace{E_{x_b}[x_s \chi_{x_s < x_b}]} \\ &= E_{x_b}[(x_b - x_s) \chi_{x_s < x_b}] + s \end{aligned}$$

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- Using Individual Rationality we sign of  $b$  and  $s$ :  
Recall that  $x_b, x_s \in [c, d]$ .

Then

- $U'_b(c) \geq 0 \Rightarrow b \geq 0$ .
- $U'_s(d) \geq 0 \Rightarrow s \geq 0$ .

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#### 4. Total Ex Ante Payoff

$$\begin{aligned} E[U'_b(x_b) + U'_s(x_s)] &= 2E[(x_b - x_s) \chi_{x_s < x_b}] + b + s \\ &> E[(x_b - x_s) \chi_{x_s < x_b}] \end{aligned}$$

Necessary payments to satisfy IR exceed the maximum total gains from trade (those prescribed by efficient outcome). QED

# Remarks

- To induce seller and buyer to reveal their information, the mechanisms must provide them certain payoffs.
- To satisfy IR constraints, those payoffs are high because IR requires a minimum payoff for certain types.
- An ex post efficient mechanism cannot get enough expected revenue to satisfy the IR constraints.
- Balance budget is not possible.

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